

Ainesh Sabharwal

Cypress, TX | aineshsab@berkeley.edu | 832-302-6497 | linkedin.com/in/ainesh-sabharwal | aineshsabharwal.me

EDUCATION

University of California, Berkeley

Bachelor of Arts in Computer Science, College of Computing, Data Science, and Society.

Relevant Coursework: Linear Algebra and Differential Equations, Computer Science 61A, Data Structures and Algorithms (Java), AP Physics C

Berkeley, CA

Expected 2030

Bridgeland High School

Class Rank 2 of 921. GPA: 4.0/4.0 unweighted.

Cypress, TX

2026

SKILLS

Languages: Java, TypeScript, Python, Swift

CS Foundations: stacks, queues, ring buffers, priority queues, trees, recursion, greedy algorithms

Web and Full-Stack: React, HTML/CSS, full-stack web applications, dashboards and analytics

AI / ML: Retrieval-Augmented Generation (RAG), on-device LLM inference, document summarization

Tools: Git, GitHub, Eclipse

EXPERIENCE

Lead Instructor

Oct 2025 – Mar 2026

iCode

- Led coding and STEM classes for K-12 students, teaching programming fundamentals through hands-on projects.
- Rebuilt the standardized curriculum with the lead director after finding the original material was above students' level.

Student Researcher

Aug 2024 – May 2025

Algoverse AI Research Program

- Conducted applied research on Retrieval-Augmented Generation (RAG) systems for grounded question answering.
- Explored retrieval, embeddings, and prompting strategies to improve answer accuracy against source documents.
- Developed AI-generated question pipelines and evaluated output faithfulness; research concluded when the project reached a stage requiring faculty graders to assess generated questions.

PROJECTS

Data Structures and Algorithms Labs github.com/sabharwalainesh

Java

- Implemented Huffman compression, a 20 Questions binary decision tree with file-backed learning, real-time audio synthesis via Karplus-Strong on a custom ring buffer, and a Java Swing image editor with undo/redo backed by a custom array-based stack.

HuddleHub

Full-Stack Local Business Platform, TypeScript, FBLA 2025-26

- Built a full-stack web application to browse, search, review, and bookmark local businesses, including an owner dashboard to manage listings and track engagement analytics.

Bridgeland Arena Hub

Event Ticketing and Box Office Platform, React/TypeScript

- Built a ticketing platform for an event venue covering event browsing, seat selection, purchase flow, and a staff-side dashboard for managing events and tracking sales.

ApathyAI github.com/qurashisohaib/ApathyAI

On-Device AI Copilot, Python, Contributor

- Contributed to an open-source desktop AI assistant that runs fully on-device; implemented on-screen context reading to auto-summarize documents and suggest spreadsheet and form fills.

AWARDS & HONORS

- Class Rank 2 of 921, Bridgeland High School
- 3rd Place, Website Design, FBLA National Leadership Conference (2025)
- Eagle Scout, Boy Scouts of America
- Black Belt, Taekwondo